

# Unit 2.6 Derivatives and rate of

Note Title

2012/03/22

change

Section 2.7 pp 143 - 148

Def The derivative of a function  $f$  at  $x=a$  is denoted by  $f'(a)$  is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

if the limit exists

OR

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

- $f'(a)$  is the rate of change of  $y = f(x)$  with respect to  $x$  when  $x = a$ .

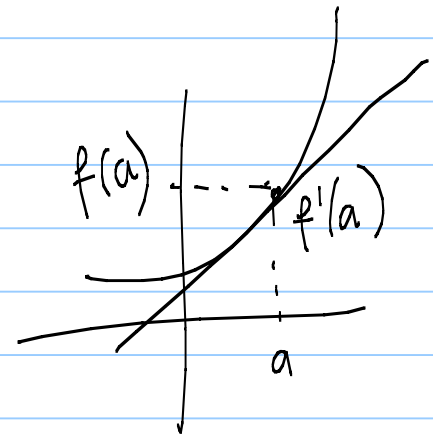
The tangent line to the curve  $y = f(x)$  at the point  $P(a, f(a))$  is the line

through the point  $P$  with a slope

$$m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

$$= f'(a).$$



The equation of a tangent line  
at the point  $a$  is:

$$(y - f(a)) = f'(a)(x - a)$$

$$(y - y_0) = m(x - x_0)$$

Example Find the equation of a tangent line to the parabola  $y = x^2$  at the point  $P(1,1)$

$$m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$= \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1}$$

$$= \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{(x-1)}$$

$$= \lim_{x \rightarrow 1} x+1 = 2$$

$$(y - f(1)) = f'(1) (x - 1)$$

$$\Rightarrow y - 1 = 2(x - 1)$$

$$\Rightarrow y = 2x - 1$$

## Homework

Find  $f'(1)$  and the equation of the tangent line if

$$f(x) = \frac{1}{x+2}$$

Example Find  $f'(2)$  using  
a table of values

|        |   |     |     |     |
|--------|---|-----|-----|-----|
| $x$    | 1 | 1.5 | 2   | 3   |
| $f(x)$ | 1 | 1.1 | 1.4 | 1.8 |

$$\frac{f(2) - f(1.5)}{2 - 1.5} = \frac{1.4 - 1.1}{0.5} = 0.6$$

$$\frac{f(3) - f(2)}{3 - 2} = \frac{1.8 - 1.4}{1} = 0.4$$

$$f'(2) \approx \frac{0.6 + 0.4}{2} = 0.5.$$

